SANDEVI DE SILVA

Software Engineering Undergraduate • Full-Stack & Systems Developer

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PROFESSIONAL PROFILE

Third-year B.Sc. (Hons) Information Technology & Management undergraduate at the University of Moratuwa with a 3.61 CGPA (Second Class Upper), actively seeking a Software Engineering Internship. Demonstrated engineering experience across web and mobile platforms with TypeScript, React, React Native, Node.js, Express, PostgreSQL, and MongoDB. Strong competencies in architecting resilient RESTful APIs, AST-based code analysis, real-time WebRTC/Socket.io systems, secure RBAC/JWT access control, and automated test-driven deployment.

EDUCATION

University of Moratuwa
B.Sc. (Hons) in Information Technology & Management
Moratuwa, Sri Lanka | 2024 – 2028
CGPA: 3.61 / 4.00 • Second Class Upper

ESOFT Metro Campus
Diploma in Information Technology
Sri Lanka | 2022 – 2023
Distinction Pass

Kegalle Balika Vidyalaya
G.C.E. Advanced Level – Physical Science Stream
Kegalle, Sri Lanka | 2020 – 2022
Combined Maths, Physics, Chemistry

TECHNICAL SKILLS

LANGUAGES	TypeScript, JavaScript (ES6+), PHP, SQL, Python, C, Java
FRONTEND & MOBILE	React 18, React Native (Expo), Tailwind CSS, HTML5, CSS3, Responsive Design
BACKEND & REALTIME	Node.js, Express.js, Socket.io, WebRTC (P2P), RESTful API Architecture, OpenAPI 3.0/3.1
DATABASES & ORM	PostgreSQL, MySQL, MongoDB, Prisma ORM, Firebase Realtime/Firestore
DEVOPS & TESTING	Docker, Git, GitHub, Jest, Supertest, Vitest, Postman, CI/CD Fundamentals
SECURITY & SERVICES	JWT, Role-Based Access Control (RBAC), SSRF Protection, Cloudinary, Google Maps API

FEATURED PROJECTS

APIHub – Full-Stack API Testing & Documentation Platform `IN ACTIVE DEVELOPMENT`

Repository →

TypeScript • React 18 • Node.js • Express.js • Prisma ORM • PostgreSQL • Tailwind CSS • Docker • OpenAPI 3.1

- Architecting an enterprise developer portal for OpenAPI 3.0/3.1 ingestion, interactive live request execution, and automated schema-driven API documentation generation.
- Engineered an SSRF-hardened backend proxy runner supporting environment variable interpolation, header injections, and multi-protocol auth (Bearer, Custom API Key, Basic).
- Built an AST-based static codebase analyzer capable of reverse-engineering Express and REST route signatures, middleware chains, and parameters without running uploaded code.
- Implemented fine-grained project-level RBAC (Owner, Editor, Viewer) backed by cryptographically secure JWT authentication and user session management.
- Integrated execution telemetry analyzing HTTP status code distributions, latency percentiles, endpoint health, and historical execution logging.
- Developed AI-assisted endpoint specification and boilerplate code generation for Express, FastAPI, NestJS, and Spring Boot.

FEATURED PROJECTS (CONTINUED)

StrayCare – Mobile Stray Animal Rescue & Community Platform

Repository →

TypeScript • React Native (Expo) • Node.js • Express.js • MongoDB • Socket.io • WebRTC • Firebase • Cloudinary • Google Maps API

- Engineered a production-ready mobile platform uniting citizens, rescuers, and animal clinics for localized emergency reporting, pet adoption, and lost-and-found matching.
- Implemented geolocation tracking and Google Maps API clustering for proximity-based incident dispatches and automated notification routing to nearby volunteer responders.
- Built real-time peer-to-peer video/audio calling utilizing WebRTC signaling protocols alongside Socket.io instant messaging and Firebase push alerts.
- Designed secure multi-tier authentication using Firebase Auth and JWT, supplemented by operational RBAC for public users, field rescuers, and clinic admins.

SpendWise – Personal Finance & Budget Management System

Repository →

TypeScript • React • Node.js • Express.js • PostgreSQL • Prisma ORM • Tailwind CSS • Recharts • Jest • Supertest

- Constructed a financial tracking application facilitating multi-account income/expense monitoring, granular budgeting thresholds, and automated savings progress trackers.
- Engineered resilient REST endpoints protected by JWT session validation, ensuring tenant isolation and user-scoped data security.
- Built interactive analytical dashboards with Recharts for visual breakdowns of monthly spending patterns, budget drift, and category velocities.
- Authored automated integration and API unit test suites utilizing Jest and Supertest to validate transaction integrity and edge-case handling.

BlogNet – MVC Multi-User Publishing & Community Portal

Repository →

PHP • MySQL • JavaScript (ES6) • Apache • HTML5/CSS3 • Markdown Engine • Git

- Developed a modular full-stack web application following Model-View-Controller (MVC) architectural principles with pure object-oriented PHP and MySQL.
- Implemented secure authentication workflows featuring password hashing, anti-CSRF token validation, and multi-tier role permissions (Admin, Author, Subscriber).
- Created rich content publishing workflows with Markdown rendering, media asset handling, and full-text keyword indexing.

COMPETITIVE PROGRAMMING & HACKATHONS

MoraXtreme 2025
Competitive Programming & Algorithm Hackathon • University of Moratuwa

JapuraXtreme 2025
Inter-University Algorithmic Coding Competition • University of Sri Jayewardenepura

CERTIFICATIONS

AI Fundamentals
DataCamp • Credential Verified

Building RESTful APIs with Node.js & Express
LinkedIn Learning • REST & Middleware Architecture

Express Essentials & Web Apps with Node.js
LinkedIn Learning • Server Architecture & Scalability

Frontend Developer Career Path
Scrimba • Advanced React & Modern UI/UX

ACADEMIC REFERENCES

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